Inteligência Artificial - Uma viagem no tempo desde as ideias de Alan Turing até aos grandes modelos de linguagem (LLM)

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Biography - a few words about me

- Associate Professor, Lisbon School of Engineering, Polytechnic University of Lisbon
- PhD in Electrical and Computer Engineering by the University of Lisbon (IST)
- Research interests: artificial intelligence, biometrics, digital signal processing, machine learning, and pattern recognition

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Outline

1. Artificial Intelligence (AI)
2. Machine Learning (ML)
3. ML approaches
   - Supervised Learning
   - Unsupervised Learning
   - Deep Learning (DL)
4. Applications
5. Some of my own work

AI - intelligent behavior (1)

First record of intelligent behavior by a computer/program:

- 1950, Alan Turing publishes the article "Computing Machinery and Intelligence" which begins with the phrase *"I propose to consider the question Can machines think?"*
- He defined the test of intelligent behavior which he called "the imitation game"
- The "imitation game" is known as **the Turing test**
- The purpose of the test is to assess the ability of a machine to exhibit intelligent behavior in a manner equivalent to or indistinguishable from that of a human being

Alan Turing, "Computing Machinery and Intelligence", Mind, LIX, October 1950, (236): 433-460, doi:10.1093/mind/LIX.236.433, ISSN 0026-4423

AI - intelligent behavior (2)

A. M. Turing (1950) Computing Machinery and Intelligence. *Mind 49:* 433-460.

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. The Imitation Game

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous, If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

AI - intelligent behavior (3)

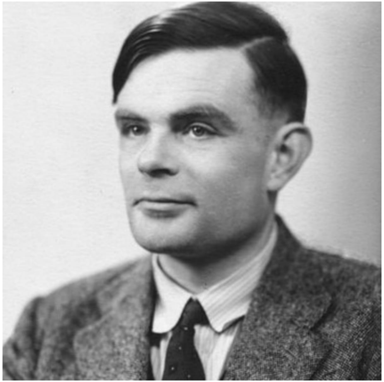
Last paragraphs of Turing's paper

We may hope that machines will eventually compete with men in all purely intellectual fields. But which are the best ones to start with? Even this is a difficult decision. Many people think that a very abstract activity, like the playing of chess, would be best. It can also be maintained that it is best to provide the machine with the best sense organs that money can buy, and then teach it to understand and speak English. This process could follow the normal teaching of a child. Things would be pointed out and named, etc. Again I do not know what the right answer is, but I think both approaches should be tried.

We can only see a short distance ahead, but we can see plenty there that needs to be done.

Alan Turing

- Alan Turing, English mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist. Known as the “father” of
  - theoretical computer science
  - artificial intelligence

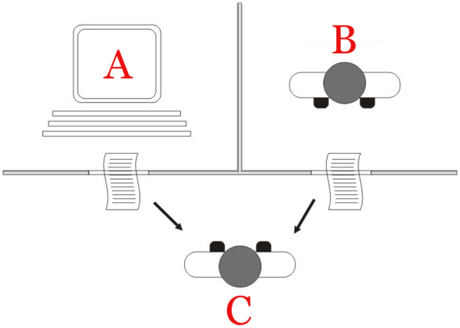


Known for	Cryptanalysis of the Enigma Turing's proof Turing machine Turing test Unorganised machine Turing pattern Turing reduction "The Chemical Basis of Morphogenesis"

Alan Mathinson Turing (1912-1954)

The Turing (intelligent behavior) test

- Human interrogator C asks questions to A and B (who are hidden from C)
- A and B answer the same question in writing
- Among A and B, one of them is human and the other is a machine
- The machine passes the test if interrogator C cannot distinguish which one is the machine
- In this case, the machine is considered to exhibit intelligent behavior and cannot be distinguished from a human being



The Imitation Game - The movie about the life of Alan Turing

- The Imitation Game
- A 2014 American historical drama film
- Directed by Morten Tyldum and written by Graham Moore, based on the 1983 biography Alan Turing: The Enigma by Andrew Hodges
- The film's title quotes the name of the game cryptanalyst Alan Turing proposed for answering the question "Can machines think?", in his 1950 seminal paper "Computing Machinery and Intelligence"

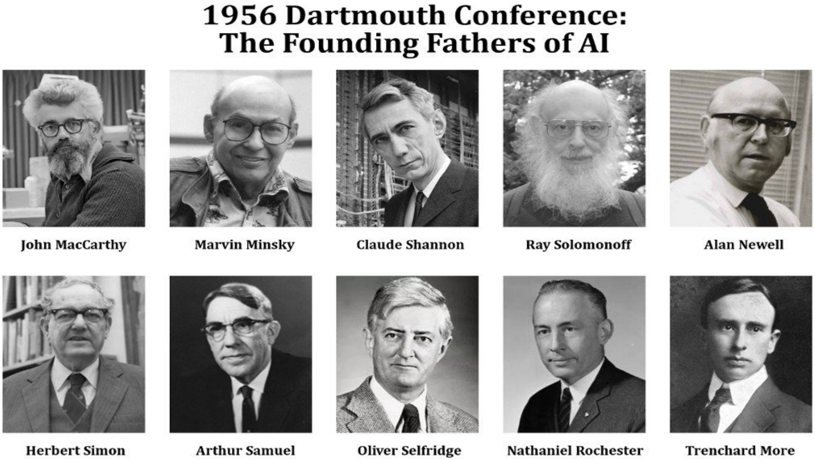
The birth of AI - the Dartmouth conference (1)

On 2 September 1955, **John McCarthy**, **Marvin Minsky**, **Nathaniel Rochester**, and **Claude Shannon** presented the following proposal to the Rockefeller Foundation, Dartmouth College (Hanover, New Hampshire, EUA)

"We propose that a 2-month, 10-man study of **artificial intelligence** *be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of* **learning or any other feature of intelligence** *can in principle be so precisely described that a* **machine can be made to simulate it***. An attempt will be made to* **find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans***, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.*

The birth of AI - the Dartmouth conference (2)

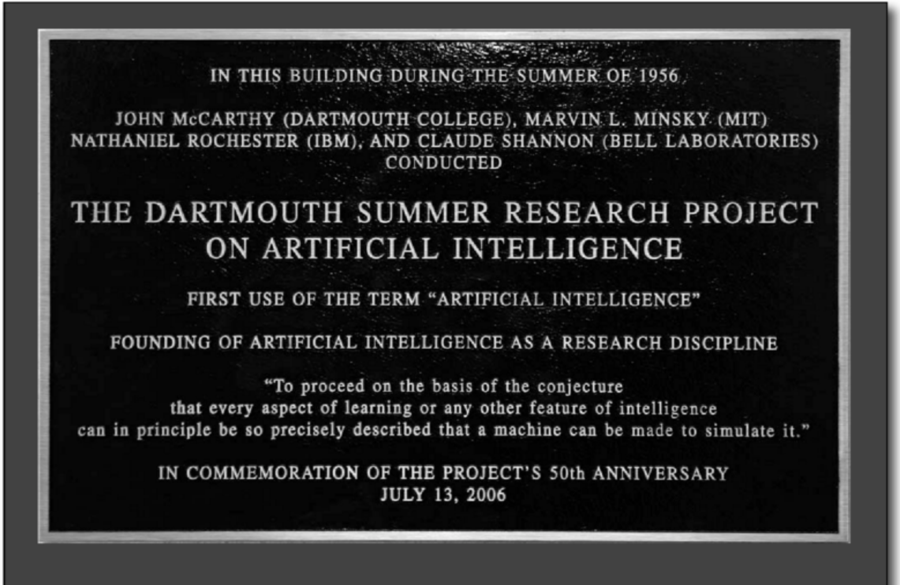
- *Dartmouth Summer Research Project on Artificial Intelligence* or *Dartmouth workshop*
- Eight weeks, on July and August 1956, with about 20 participants



50 years of AI - AI@50 conference (1)

- In 2006, we had 50 years over the Dartmouth conference
- **AI@50** conference, between 13 and 15 July 2006
- Organized by James Moor, professor at Dartmouth
- Attended by five of the 1956 participants: Trenchard More, John McCarthy, Marvin Minsky, Oliver Selfridge, and Ray Solomonoff

50 years of AI - AI@50 conference (2)



50 years of AI - AI@50 conference (3)

The goals of AI@50:

- Analyze progress on the original AI challenges over the first 50 years and assess whether the challenges were easier or harder than originally thought and why
- Document what AI@50 participants believe are the major research and development challenges of the next 50 years and identify advances needed to address these challenges
- Relate these challenges and advances to developments and trends in other areas, such as control theory, signal processing, information theory, statistics, and optimization theory

James Moore, "The Dartmouth College Artificial Intelligence Conference: The Next Fifty Years", AI Magazine, volume 27, number 4, 2006, *Association for the Advancement of Artificial Intelligence* (AAAI) `https://www.researchgate.net/publication/220605256_The_Dartmouth_College_Artificial_Intelligence_Conference_The_Next_Fifty_Years`

50 years of AI - AI@50 conference (4)

AI Magazine Volume 27 Number 4 (2006) (© AAAI)

The Dartmouth College Artificial Intelligence Conference: The Next Fifty Years

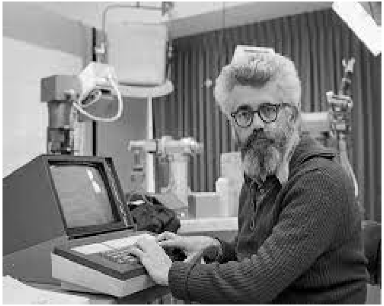
James Moor

continued this earlier work because he became convinced that advances could be made with other approaches using computers. Minsky expressed the concern that too many in AI today try to do what is popular and publish only successes. He argued that AI can never be a science until it publishes what fails as well as what succeeds.

Oliver Selfridge highlighted the importance of many related areas of research before and after the 1956 summer project that helped to propel AI as a field. The development of improved languages and machines was essential. He offered tribute to many early pioneering activities such as J. C. R. Lickleiter developing time-sharing, Nat Rochester designing IBM computers, and Frank Rosenblatt working with perceptrons.

AI - John McCarthy (1)

- John McCarthy coined the term Artificial Intelligence (AI)
- Winner of the Turing Award, in 1971 (the "Nobel" prize for Computer Science)
- Assigned by the **Association for Computing Machinery (ACM)**, founded in 1947, `https://amturing.acm.org/byyear.cfm`



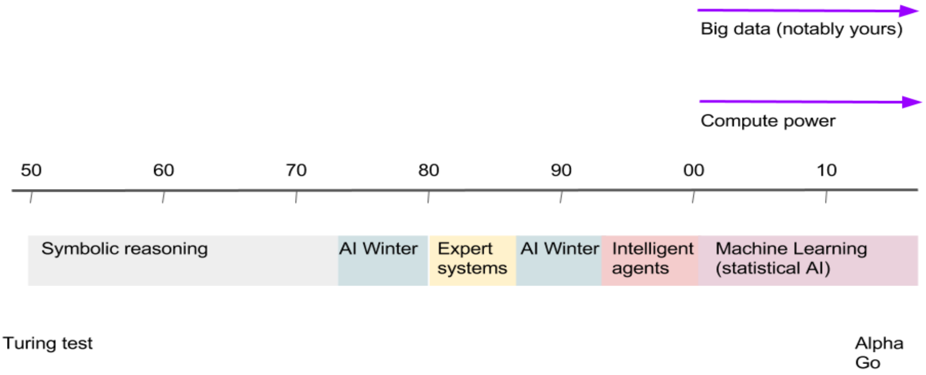
John McCarthy (1927-2011)

AI - John McCarthy (2)

John McCarthy contributions:

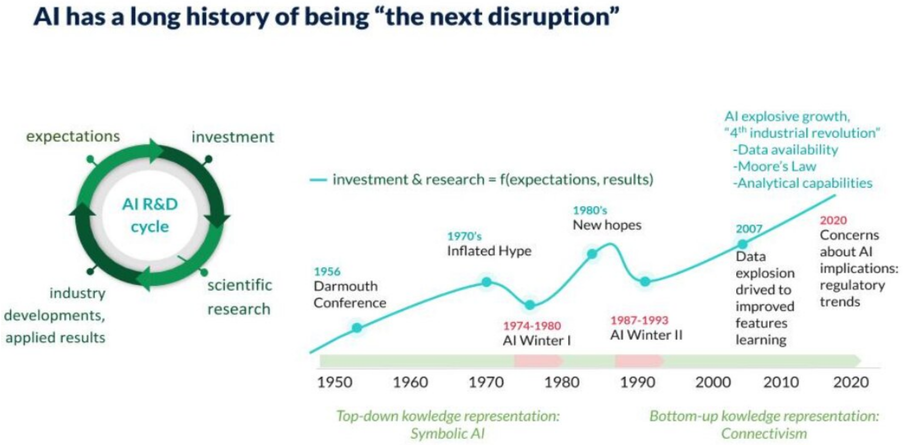
- Coined and Defined Artificial Intelligence
- Creator of the LISP Programming Language (functional paradigm)
- Early Vision of Cloud Computing (computing as a service like electricity)
- Time-Sharing Systems (multi-tasking and multi-processing)
- Gargabe Collection for programming languages (for LISP, first) - automatic memory management
- Formalization of AI Concepts
- Circumscription (Commonsense Reasoning)

AI - Timeline (1)



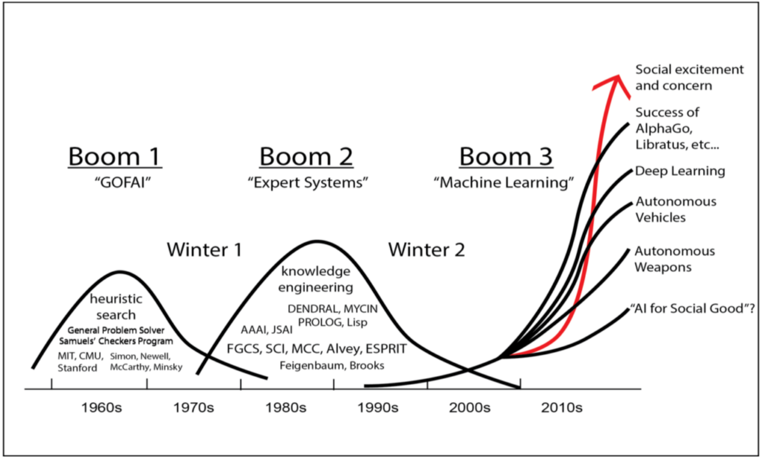
https://dpereirapaz.medium.com/is-a-new-ai-winter-coming-maybe-it-is-ai-spring-time-instead-b28a26beecb6

AI - Timeline (2)



https://www.bbva.com/en/opinion/what-should-be-taken-into-account-if-artificial-intelligence-is-to-be-regulated

AI - Timeline (3)



https://www.technologystories.org/ai-evolution

(GOFAI = Good and Old Fashioned Artificial Intelligence)

Machine Learning - Arthur Samuel (1)

- In 1959, Arthur Samuel, American pioneer in the field of computer gaming and artificial intelligence, coined the term **Machine Learning (ML)**
- In the 1950's, he developed the *Game of Checkers* program; probably the first computer program with ML; Known as the Samuel Checkers-playing program

Arthur Lee Samuel (1901 - 1990)

Arthur Samuel, "Some Studies in Machine Learning Using the Game of Checkers". IBM Journal of Research and Development. 44: 206-–226.

Machine Learning - Arthur Samuel (2)

Some Studies in Machine Learning Using the Game of Checkers

Arthur L. Samuel

Abstract: Two machine-learning procedures have been investigated in some detail using the game of checkers. Enough work has been done to verify the fact that a computer can be programmed so that it will learn to play a better game of checkers than can be played by the person who wrote the program. Furthermore, it can learn to do this in a remarkably short period of time (8 or 10 hours of machine-playing time) when given only the rules of the game, a sense of direction, and a redundant and incomplete list of parameters which are thought to have something to do with the game, but whose correct signs and relative weights are unknown and unspecified. The principles of machine learning verified by these experiments are, of course, applicable to many other situations.

Originally published in *IBM Journal*, Vol. 3, No. 3, July, 1959.

Machine Learning - Arthur Samuel (3)

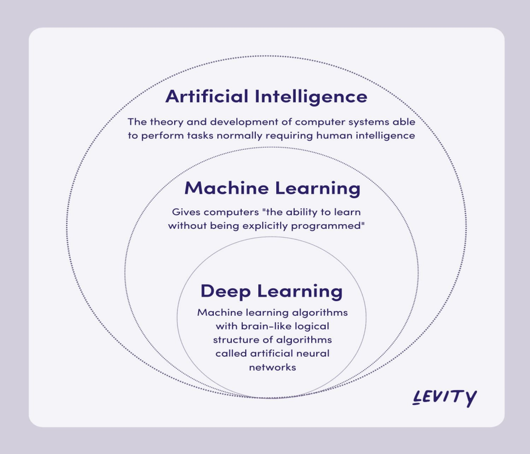
Arthur Samuel's other contributions:

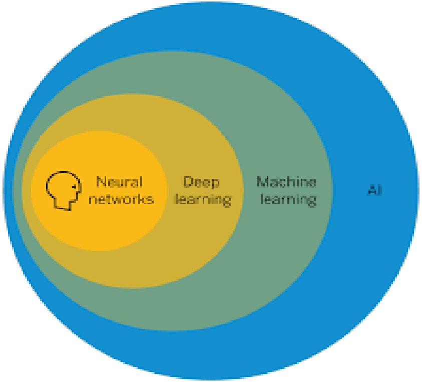
- Alpha–beta pruning AI algorithm, early implementation
- Tex* typesetting system project, with Donald Knuth (Turing award, 1974); author of the Tex manual in 1983
- Tex provided the ability to write articles that explain complex topics in a simple way
- In 1985, Leslie Lamport (Turing award, 2013) developed LaTeX (a document preparation system), an extension of Tex (a typesetting program); LaTeX = Lamport + TeX
- His work influenced the instruction set of IBM computers `https://en.wikipedia.org/wiki/IBM_701`
- TeX, derived from ancient Greek which means "art", "skill", "technique"

Arthur Lee Samuel	
Born	December 5, 1901 Emporia, Kansas
Died	July 29, 1990 (aged 88) Stanford, California
Citizenship	United States
Alma mater	MIT (Master 1926) College of Emporia (1923)
Known for	Samuel Checkers-playing Program Alpha–beta pruning (an early implementation) Pioneer in Machine Learning[2] TeX project (with Donald Knuth)
Awards	Computer Pioneer Award (1987) [1]
Scientific career	
Fields	Computer Science
Institutions	Bell Laboratories (1928) University of Illinois (1946) IBM Poughkeepsie Laboratory (1949) Stanford University (1966)

ML and AI

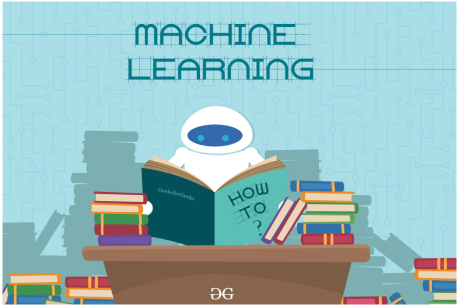
- ML is a subset of AI
- Deep Learning (DL) is a subset of ML
- DL resorts to the use of Neural Networks (NN)

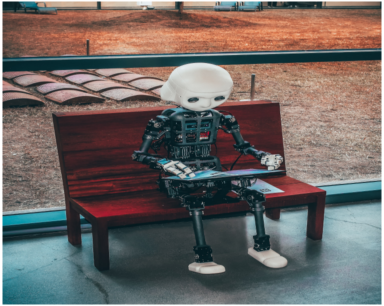




ML - Definition (1)

Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed (Arthur Samuel)



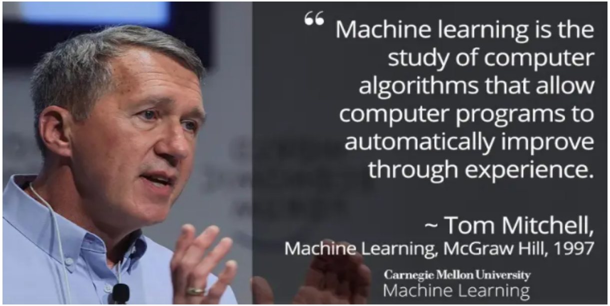


`https://www.geeksforgeeks.org/machine-learning`

`https://pacificsun.com/from-ba-to-ai-students-seek-careers-in-artificial-intelligence`

ML - Definition (2)

Machine learning definition *"A computer program is said to learn from experience E with respect to some set of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."*



https://towardsai.net/p/machine-learning/what-is-machine-learning-ml-b58162f97ec7

ML and traditional programming (1)

Traditional programming

- To solve a problem with a computer program, we need an **algorithm**
- Problem 1 - sort the words in an email message
- Problem 2 - classify (the same) email message as being SPAM or not

To solve Problem 1, one possible **algorithm** is:

(1) read the file contents
(2) isolate the words
(3) insert the words into a data structure
(4) apply the sorting operation on that data structure

To solve Problem 2, what is the **algorithm**? What are the steps to define that **algorithm**?

Another problem: facial recognition (easy for humans, hard for the computer)

ML and traditional programming (2)

ML uses concepts and techniques from:

- mathematics (mostly, statistics and linear algebra)
- information theory
- computer science
- programming

Instead of trying to devise directly the appropriate algorithm, ML uses data (many examples) to compensate for the lack of knowledge about the exact algorithm to solve a problem

We gather a (large and representative) set of email messages labeled by a human supervisor as SPAM or not SPAM, i.e....

$\rightarrow$ **we build a (dataset)!**

ML and traditional programming (3)

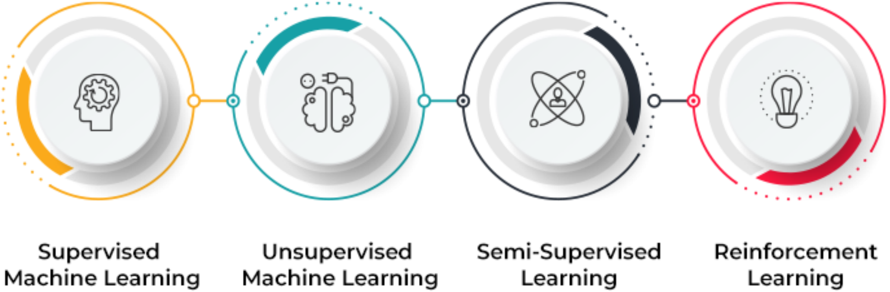
From a dataset and a **training process**, **we learn** a **model** (rule) for the classification of the email message

Example of such a model (rule):

```
if ( percentage-of-occurrence('george') < 0.15 AND
percentage-of-occurrence('you') > 0.25) then
label = 'SPAM'
else
label = 'NOT SPAM'
```

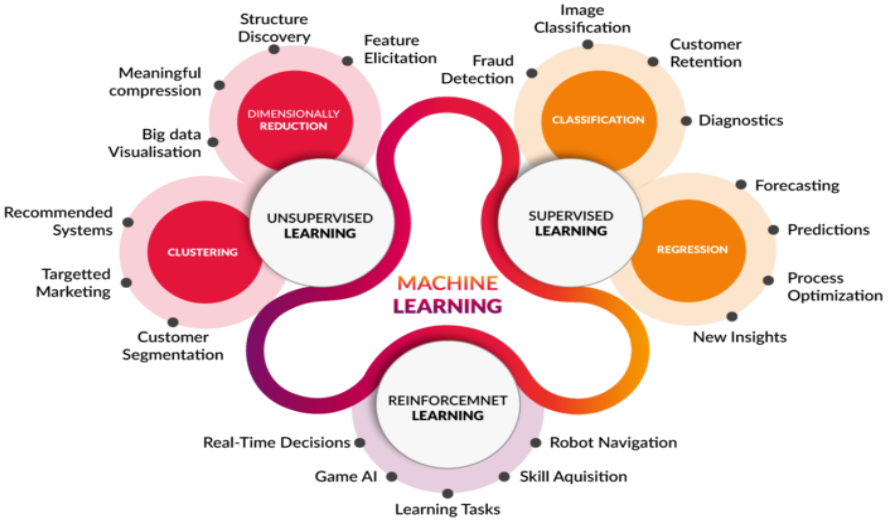
- The classification rule is a (small) program learned from the data
- This classifier must have coverage over the training set

The four typical ML approaches



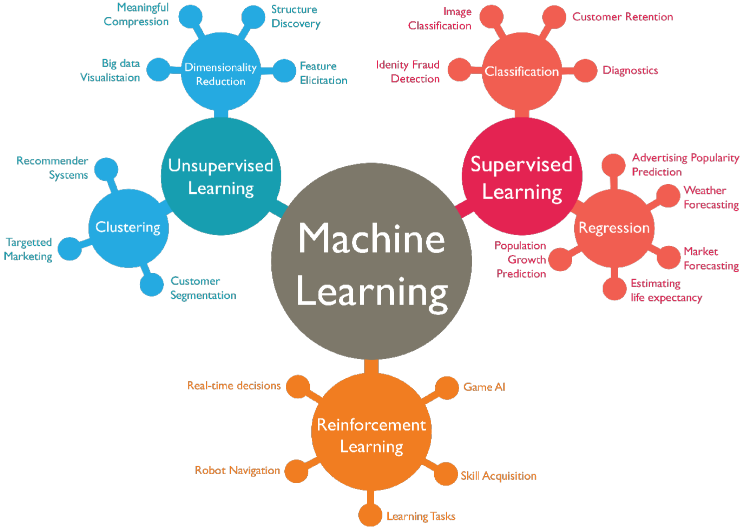
ML approaches (2)

The three typical ML approaches and their common use



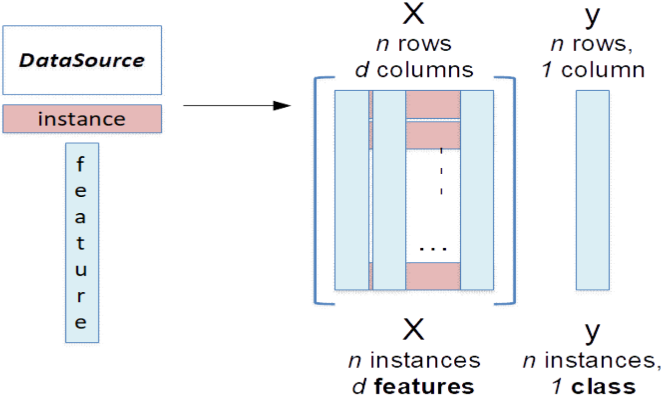
ML approaches (3)

The three typical ML approaches and their common use



Supervised Learning (1)

- **Classification** and **Regression** tasks
- Dataset = Data Matrix + Label Vector
- Data Matrix: $\mathbf{X}$, matrix of size $n \times d$
- Label Vector: $\mathbf{y}$, vector of size $n \times 1$



Supervised Learning (2)

Example of a dataset

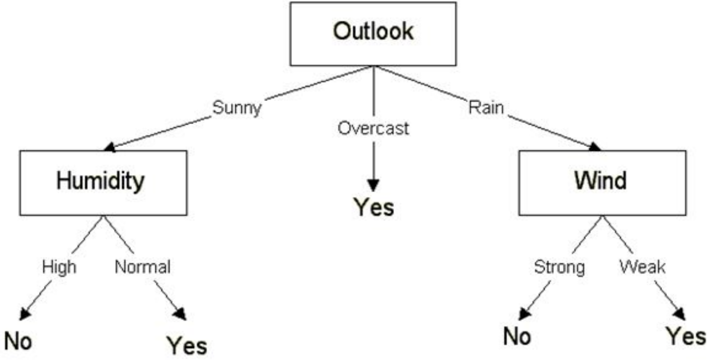
Outlook	Temperature	Humidity	Windy	Play
Sunny	85	85	False	No
Sunny	80	90	True	No
Overcast	83	86	False	Yes
Rainy	70	96	False	Yes
Rainy	68	80	False	Yes
Rainy	65	70	True	No

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes

- The problem is to should we go out to play tennis, depending on the weather
- The **play** column is the label vector, named as **y**
- The remaining columns are the data matrix, named as **X**
- This is a binary classification problem

Supervised Learning (3)

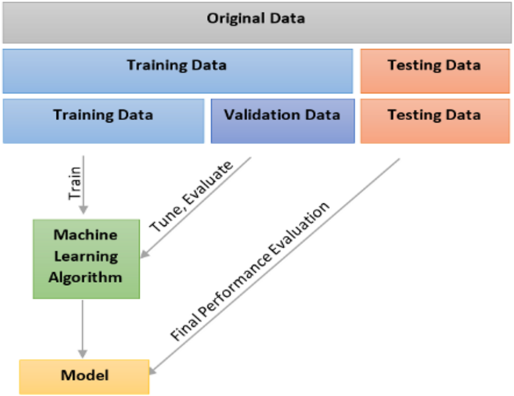
Learning result of a decision tree classifier



https://medium.datadriveninvestor.com/decision-tree-algorithm-with-hands-on-example-e6c2afb40d38

Supervised Learning (4)

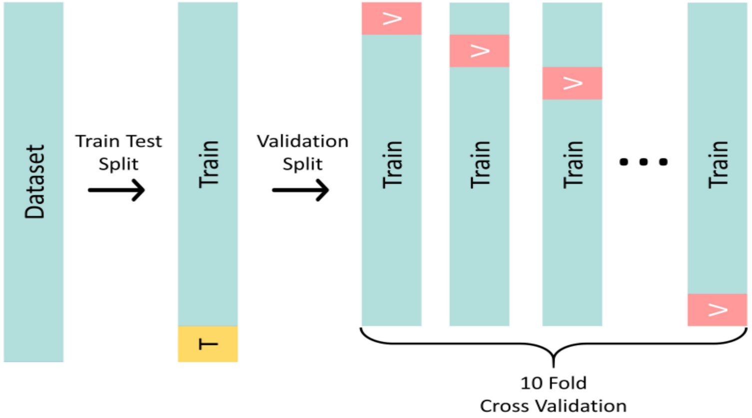
The dataset is partitioned into: **training subset**, **validation subset**, and **test set**



`https://medium.com/analytics-vidhya`

Supervised Learning (5)

For **model learning** and **model evaluation**, typically we resort to 10-fold cross-validation



https://pdfs.semanticscholar.org/75fc/2fbeb3a1e16b55a5addc76dc79df5e3e0235.pdf

Supervised Learning (6)

The three disjoint sets for **training**, **validation**, and **test**

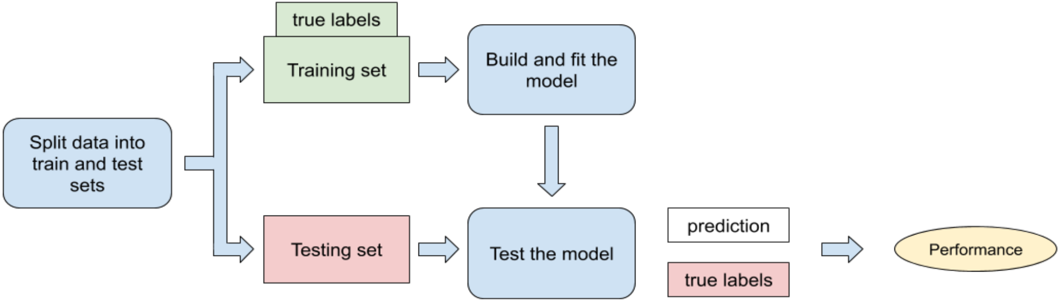
- Training set - to perform model learning
  - classifier (the target is a set of discrete labels)
  - regressor (the target is a continuous function)

- Validation set - to adjust some parameters of the model (optional), still during the training phase

- Test set - to evaluate the performance of the learned model
  - we evaluate its ability for generalization
  - the evaluation is carried out using different metrics (error rate, for example)

Supervised Learning (7)

Train (learning) and test (generalization) procedures



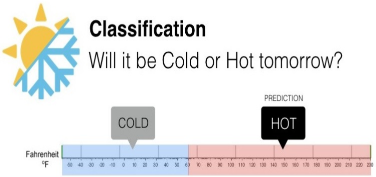
https://www3.tuhh.de/sts/hoou/data-quality-explored/0-2-supervised-learning.html

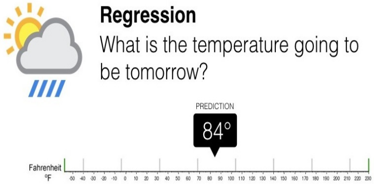
Supervised Learning (8)

Formally

- A **classifier** is a function $f : \mathbf{x} \rightarrow y, \;\; y \in \mathbb{N}$ is the discrete label
- A **regressor** is a function f, such that $f : \mathbf{x} \rightarrow y, \;\; y \in \mathbb{R}$
- Function f minimizes the **risk** or **loss**, $L(f(\mathbf{x}), \mathbf{y})$, over the training set

$\mathbf{x} = [x_1, x_2, \ldots, x_d]$ is a pattern, instance, or example
The dataset is represented by $(\mathbf{X}, \mathbf{y})$

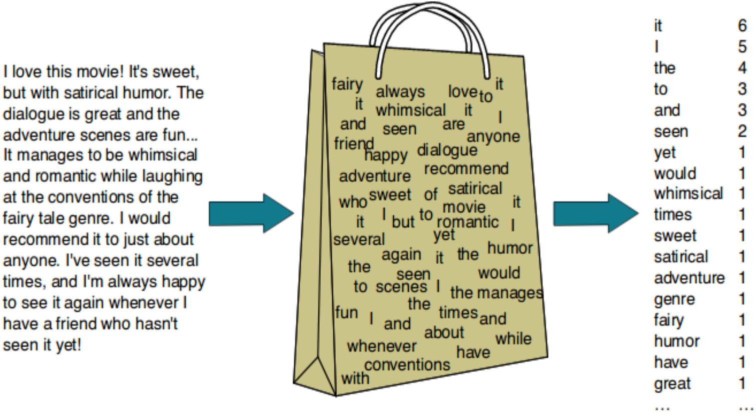




https://medium.com/@ali_88273/regression-vs-classification-87c224350d69

Supervised Learning (9)

For the problem of classifying email as SPAM or not SPAM, a representation of email messages in the form of *bag-of-words* (BoW) is typically used



https://koushik1102.medium.com/nlp-bag-of-words-and-tf-idf-explained-fd1f49dce7c4

Supervised Learning (10)

In the literature, we have a vast set of classifiers

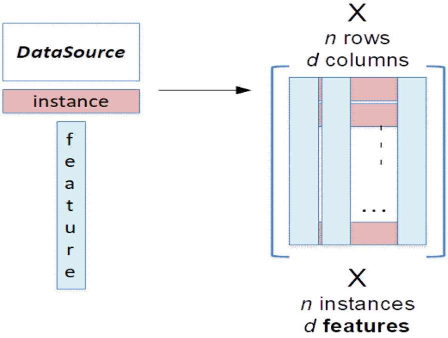
- One-rule (1R), Naïve Bayes (NB), support vector machines (SVM), decision tree (DT), k-nearest neighbor (k-NN),...
- Boosting, Adaptive Boosting (AdaBoost) and its variants,...
- Neural networks (NN), a large family of topologies

We also have many regression techniques

- Organized into three families: linear, multiple linear, and non-linear
- Linear, polynomial, support vector, decision tree, random forest, ridge, LASSO, and logistic regression

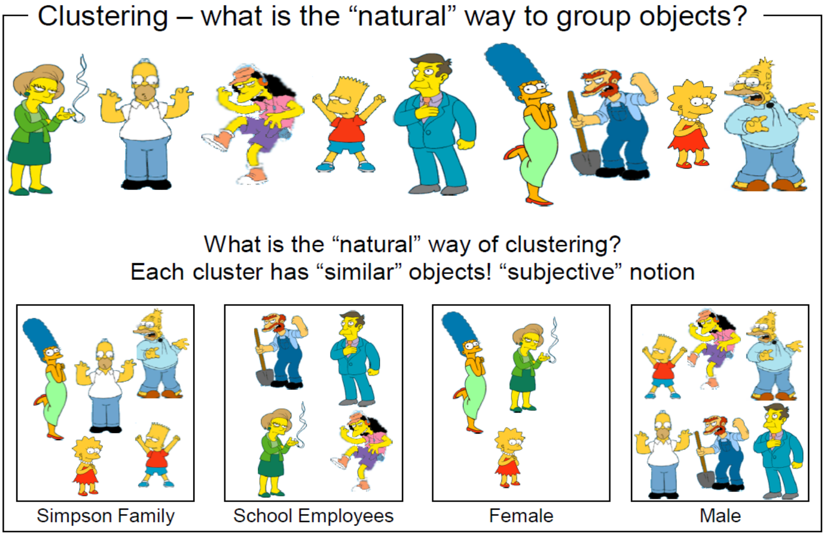
Unsupervised Learning (1)

- We have **clustering** and **association** tasks
- There is no supervisor, no data labeling nor explicit targets
- The dataset is composed by the Data Matrix, $\mathbf{X}$



- The goal is to cluster the data, according to some similarity metric
- We aim to cluster/group the instances that represent the same concept

Clustering by **similarity** or with some defined **criteria**



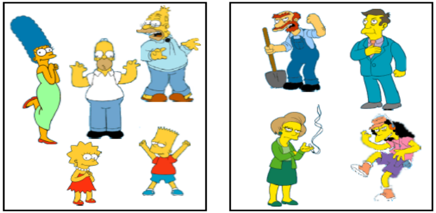
Unsupervised Learning (3) - clustering

Non-hierarchical and hierarchical clustering

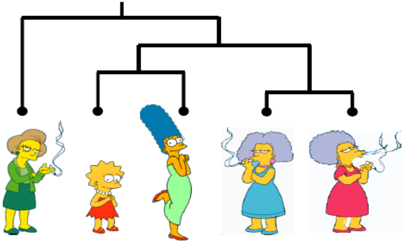
Non-Hierarchical (partition), mutually exclusive partitions of the entire object space are created

Hierarchical (decomposition), a tree is created where each level is made up of mutually exclusive partitions

Non-Hierarchical (Partition)



Hierarchical



Unsupervised Learning (4) - clustering and similarity

Unsupervised Learning $\rightarrow$ No Supervisor to the Learning Process!
Then, one must define some similarity metric

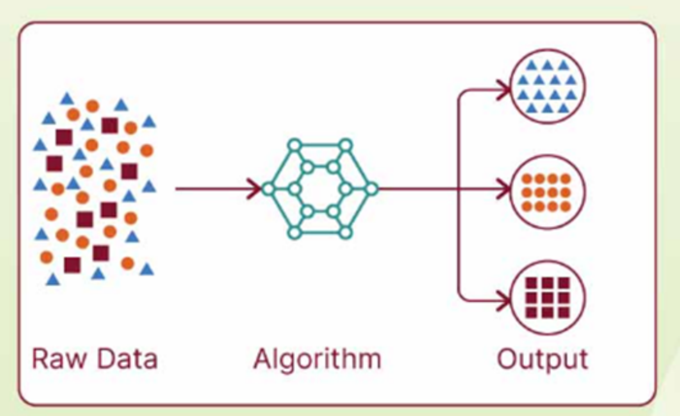
The quality or state of being similar; likeness; resemblance; as a similarity of features.

in Webster's Dictionary

Similarity is hard to define, but we all know when we encounter similar objects....

Unsupervised Learning (5) - clustering algorithms

Clustering with $k = 3$ clusters (K-Means algorithm)

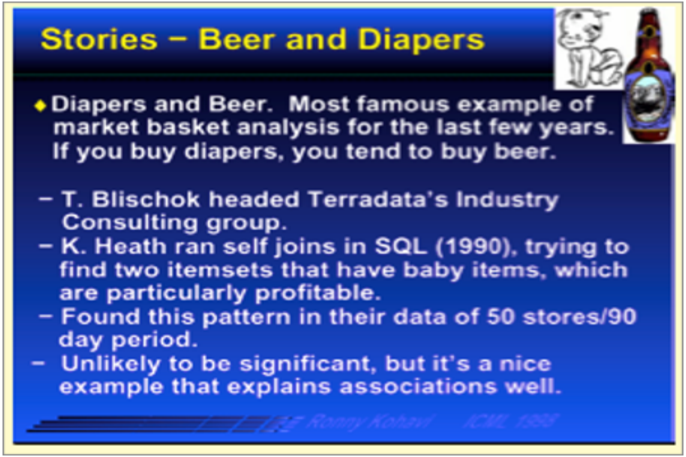


Other clustering algorithms: DBSCAN, EAC, and GMM https://www.shiksha.com/online-courses/articles/types-of-clustering-algorithm-scenario-you-must-know-as-a-data-scientist

Unsupervised Learning (6) - association rules

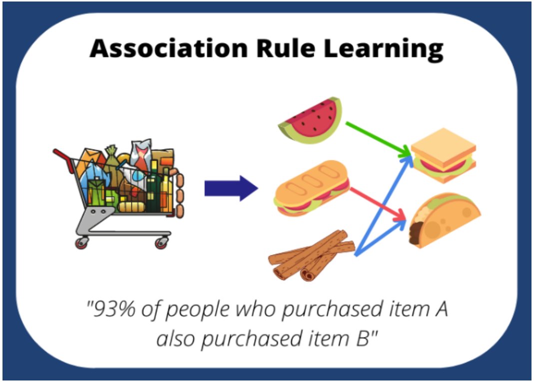
Association Rule (AR) mining is a ML technique that finds patterns in data by identifying which items are frequently bought together

The beer and diapers purchase association



Unsupervised Learning (7) - association rules

The APRIORI algorithm (Agrawal e Srikant, 1994) and the market-basket analysis

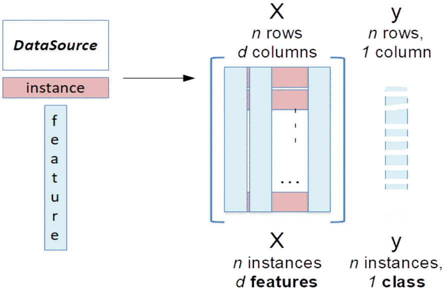


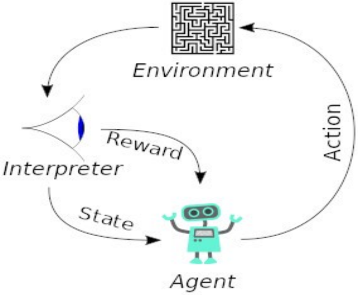
https://en.wikipedia.org/wiki/Apriori_algorithm
https://analyticsarora.com/what-is-association-rule-learning-machine-learning-interview-questions

Semi-supervised and Reinforcement Learning

The other ML forms:

- **Semi-supervised**, only a portion of the instances is labeled
- **Reinforcement learning** is based on a sequence of actions
  - the outcome of each action has a positive or negative reinforcement
  - in general, it is more important to reinforce the outcome of a sequence of actions than the individual actions

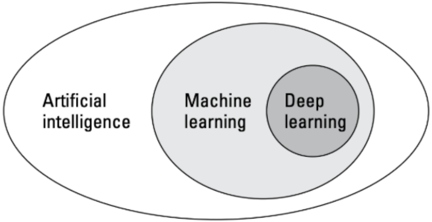




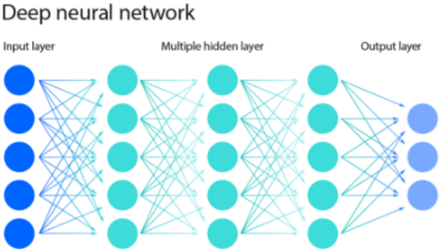
https://neptune.ai/blog/reinforcement-learning-applications

Deep Learning (DL) (1)

- DL is a subset of ML
- The “deep” in deep learning refers to the depth of layers in a neural network



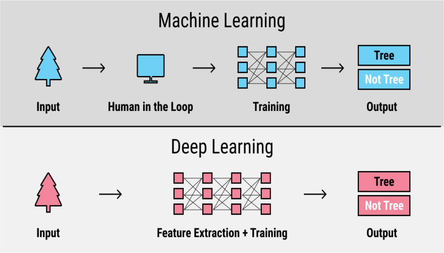
Deep learning, a subset of a subset of AI.

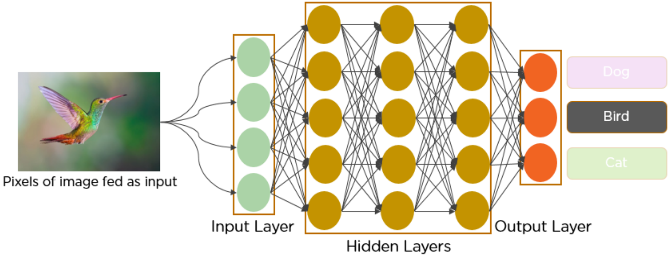


https://www.deepinstinct.com/glossary/deep-learning
https://www.ibm.com/think/topics/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks

Deep Learning (2)

- With DL, the intervention of the (human) data scientist is minimized
- The *feature engineering/extraction* phase is (very) simplified

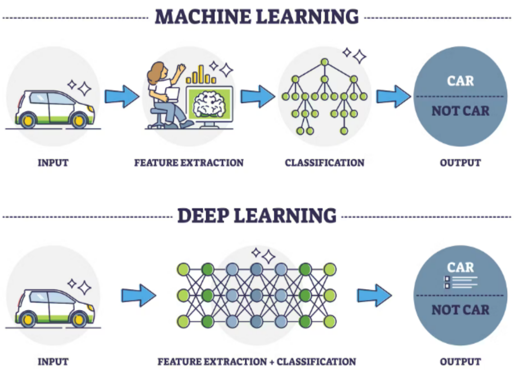




https://labelyourdata.com/articles/machine-learning-and-training-data
https://www.analyticsvidhya.com/blog/2021/05/convolutional-neural-networks-cnn

Deep Learning (3)

- The feature extraction phase is a critical step for ML algorithms success
- How many features? What features to chose?



https://www.turing.com/kb/ultimate-battle-between-deep-learning-and-machine-learning

Some applications of AI, ML, and DL

In different domains and different problems (everywhere, really):

- banking - credit scoring
- stock market - prediction and forecasting the market evolution
- smartphones (biometrics) - facial recognition, fingerprint recognition, speech recognition
- reading documents - optical character recognition (OCR)
- medicine - cancer diagnosis
- weather forecasting
- electronic commerce - costumer profile and recommender systems
- streaming platforms such as NetFlix and AmazonPrime
- social media - analysis of the multimedia contents
- self-driving cars
- quantum physics, `https://en.wikipedia.org/wiki/Machine_learning_in_physics`
- ...

Autonomous Driving - Tesla

Tesla's Autopilot and Full Self-Driving (FSD) are Advanced Driver-Assistance Systems (ADAS) that use cameras to help with driving using

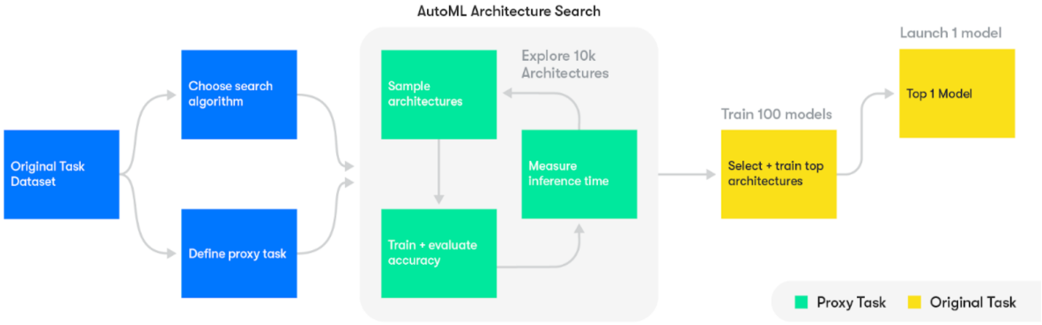
- Deep Neural Networks, namely Convolutional Neural Networks (CNN)
- Computer Vision with the **HydraNet network**
- Reinforcement Learning
- Data from 8 external video cameras and ultrasonic sensors

The **HydraNet network**:

- A shared backbone, with a CNN that processes the raw data from cameras
- Multiple heads, each one with a specific task
    - Lane Detection: Identifies road lines
    - Object Detection: Cars, pedestrians, bicycles
    - Depth Estimation: Calculates the distance of objects
    - Semantic Segmentation: Understands what is a sidewalk, street, wall, etc.
    - Traffic Sign Detection: Reads and classifies signs
    - Motion Prediction: Calculates how objects will move

Autonomous Driving - Google

Google - Waymo AutoML

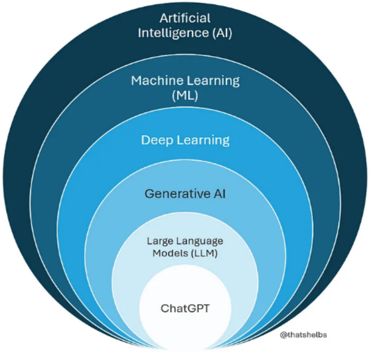


"We used a new machine learning technique called Deep Learning with help from the Google Brain team to teach our cars about the nature of objects on the road and understand how they would react" - Dmitri Dolgov, Co-CEO, Waymo

https://blog.waymo.com/2019/07/automl-automating-design-of-machine.html

ChatGPT - OpenAI (1)

- ChatGPT, Generative Pre-trained Transformer (GPT)
- Transformer is a NN model for natural language (text)
- Transformer was introduced in the famous paper *"Attention Is All You Need" (2017), by 8 Google scientists [175945 citations]*
- It resorts to the **attention mechanism** to process everything at once and not sequentially
- Large Language Model (LLM) understands structure and meaning of the text between humans and machines
- Generative AI, a type of AI that creates new content — like text, images, music, code, video, and more — that looks like it was made by a human

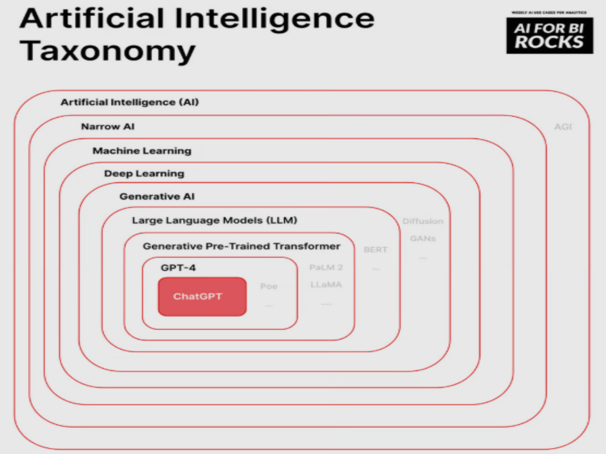


Artificial Intelligence to ChatGPT Stacked Venn Diagram

https://generativeai.pub/stop-confusing-ai-with-generative-ai-understanding-the-key-differences-21da2b2d33740

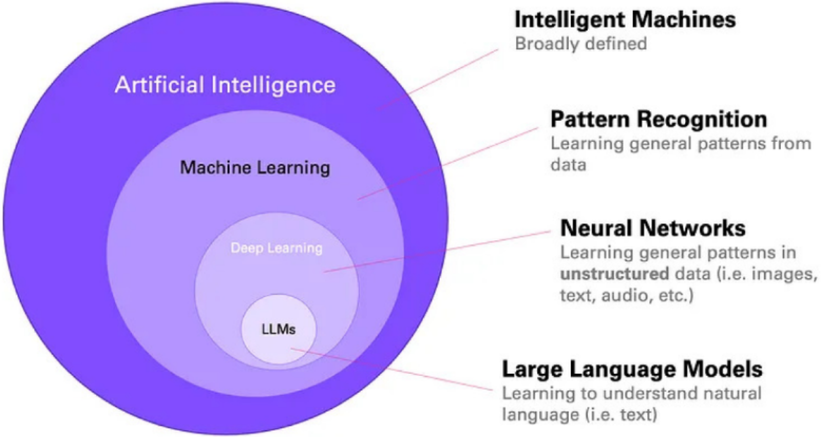
ChatGPT - OpenAI (2)

ChatGPT under the AI framework



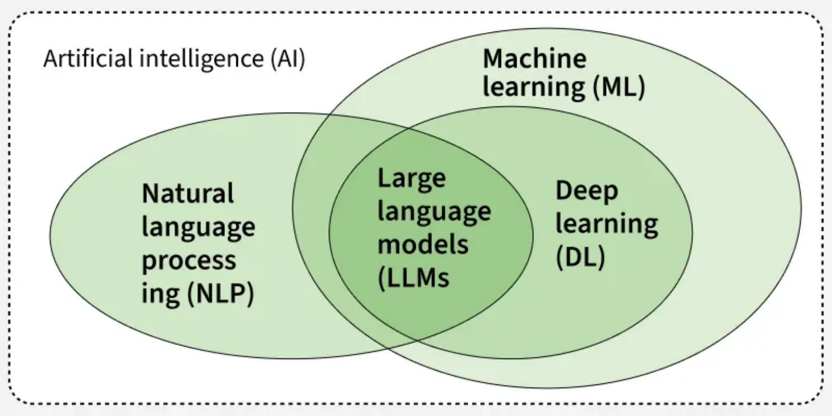
https://www.gptechblog.com/5-diagrams-to-help-you-understand-generative-ai/

Large Language Models (LLM)



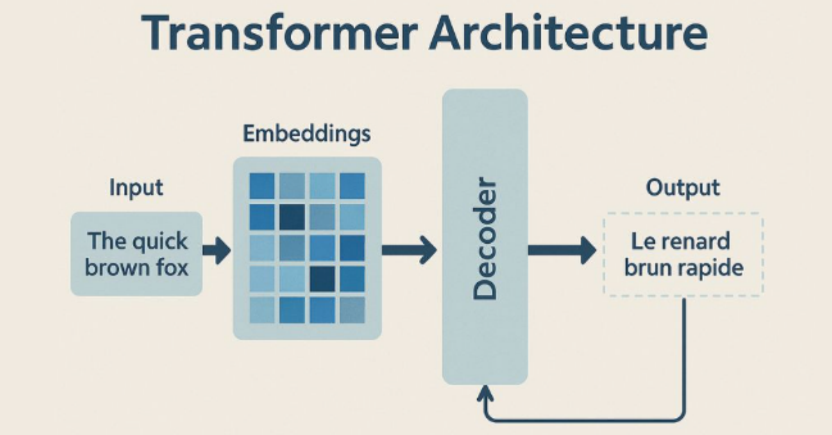
https://www.ultralytics.com/pt/blog/from-code-to-conversation-how-does-an-llm-work

Large Language Models (LLM)



https://www.geeksforgeeks.org/nlp/nlp-vs-llm/

Transformer Architecture (1)

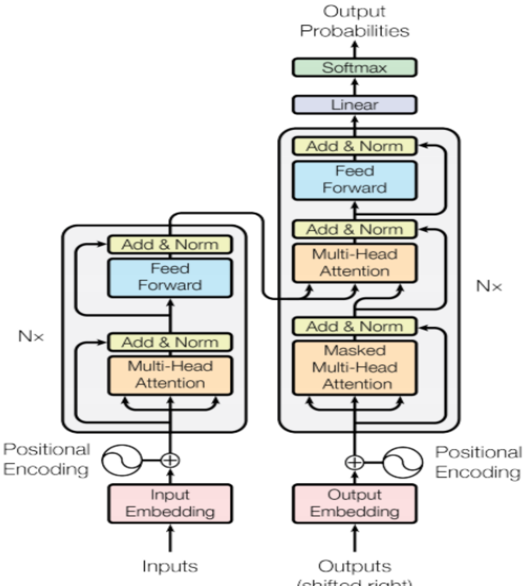


https://www.mygreatlearning.com/blog/understanding-transformer-architecture/

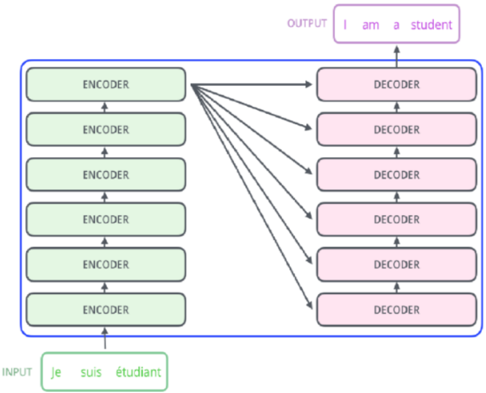
Transformer Architecture (2)

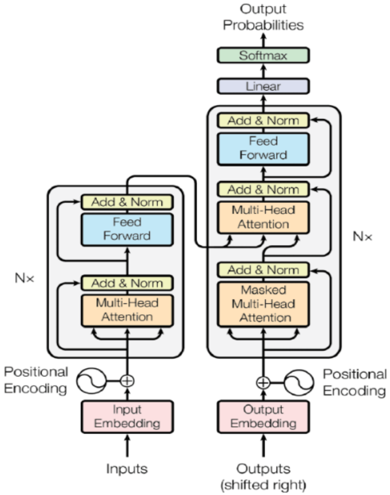
- The transformer architecture is a deep learning model introduced in the paper Attention Is All You Need by Vaswani et al. (2017)
- It eliminates the need for recurrence by using self-attention and positional encoding
- It is highly effective for sequence-to-sequence tasks such as language translation and text generation.

Transformer Architecture (3)



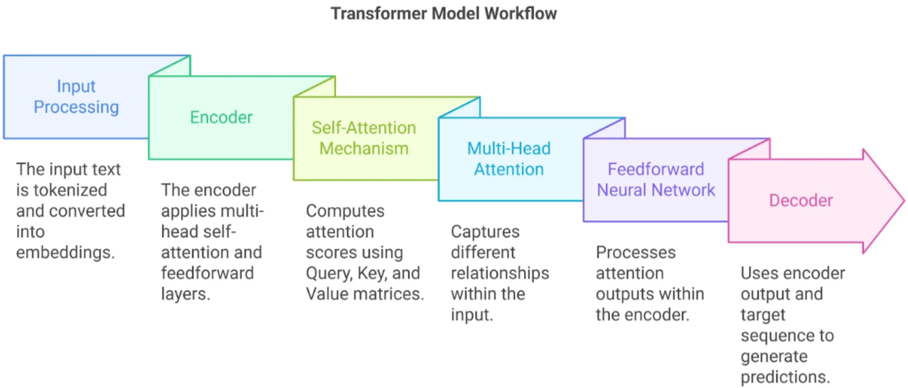
Transformer Architecture (4)





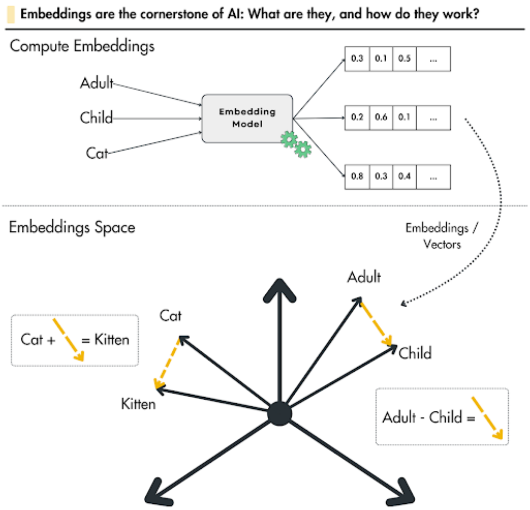
https://www.linkedin.com/pulse/transformer-model-neural-network-which-uses-attention-tejas-bankar/

Transformer Architecture (5)



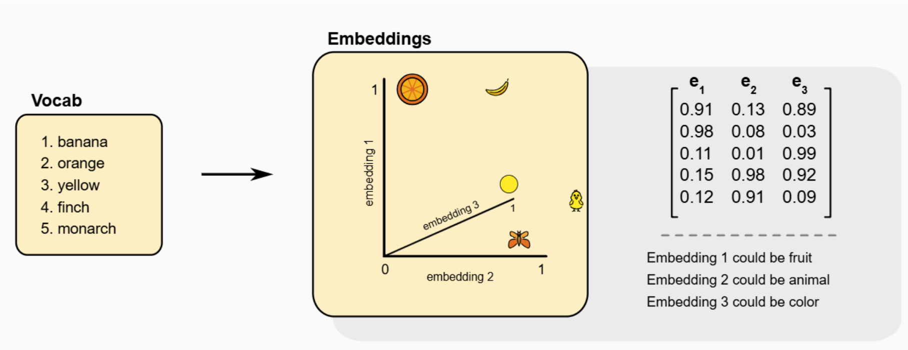
https://www.mygreatlearning.com/blog/understanding-transformer-architecture/

Embeddings (1)



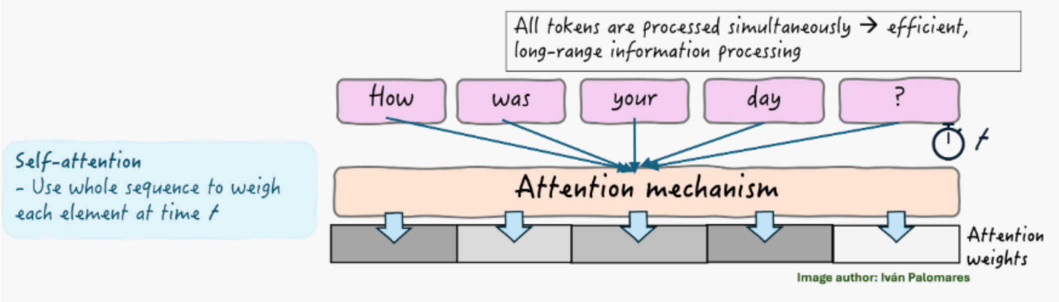
https://www.lightly.ai/blog/embeddings

Embeddings (2)



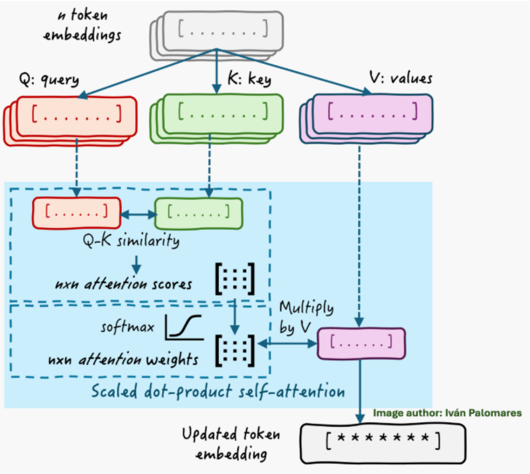
https://www.lightly.ai/blog/embeddings

Attention Mechanism (1)



https://machinelearningmastery.com/attention-may-be-all-we-need-but-why/

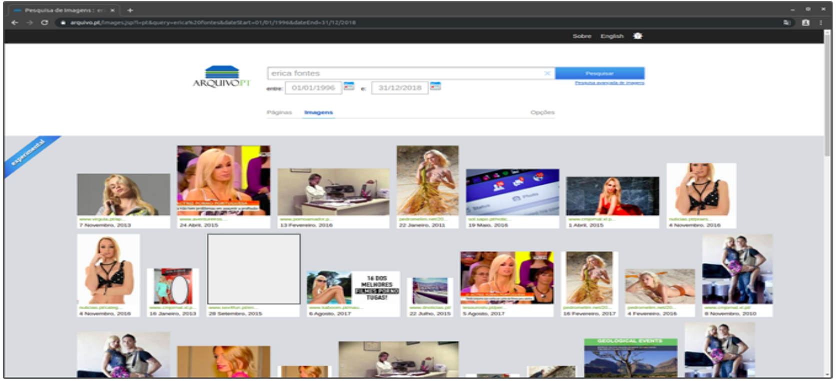
Attention Mechanism (2)



https://machinelearningmastery.com/attention-may-be-all-we-need-but-why/

Detection of *not-safe-for-work* (NSFW) images for Arquivo.pt

- Use of DL with ResNet and SqueezeNet models
- Integration with Arquivo.pt
- Access trough the browser



Driver style detection from telematics i-DREAMS data

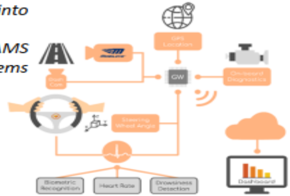
PROPOSED SOLUTION AND EXPERIMENTAL RESULTS

- An unsupervised **Machine Learning (ML)** approach
- Identify driver styles based on driver data
- Anonymized driver data from ***Mobileye*** and ***CardioWheel***

Mobileye *CardioWheel*

Integration into

The i-DREAMS on-vehicle systems



Events

System	Data Identifier
Gateway	Ignition
CardioWheel	LOD_Event
Drowsiness Detector	Drowsiness
Gateway Motion Sensor	DrivingEvents
Mobileye	ME_AWS
Mobileye	ME_TSR
Mobileye	ME_Car
GNSS	Geolocation Coordinates
Driver App	Distraction
Safety Tolerance Zone 1	iDreams_Fatigue
Safety Tolerance Zone 2	iDreams_Headway
Safety Tolerance Zone 3	iDreams_Overtaking
Safety Tolerance Zone 4	iDreams_Speeding

Features

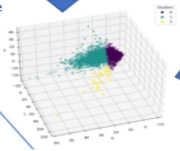
Dataset with
n = 15002 instances (trips)
d = 53 features

Clustering algorithms

- K-Means
- DBSCAN
- Gaussian mixture

First stage

Second stage



K-Means clustering metrics

Clustering Scores	Normalization by distance			Normalization by duration		
	PCA	SVD	None	PCA	SVD	None
CH ↑	7258.9	**7259.5**	7162.8	4503.6	4614.7	4563.2
DB ↓	**1.1**	**1.1**	**1.1**	**1.1**	1.3	1.3
S ↑	**0.48**	**0.48**	**0.48**	0.44	0.43	0.43
Instances cluster 0	11916	11852	11852	12367	12358	12416
Instances cluster 1	3086	3150	3150	2635	2644	2586

DBSCAN clustering metrics

Clustering Scores	Normalization by distance			Normalization by duration		
	PCA	SVD	None	PCA	SVD	None
CH ↑	2365.4	2365.6	**2913.5**	1844.4	1918.2	2149.8
DB ↓	**1.3**	**1.3**	1.4	1.4	1.4	1.5
S ↑	0.65	0.65	0.62	**0.71**	**0.71**	0.66
Instances cluster 0	452	452	680	277	293	469
Instances cluster 1	14550	14550	14322	14725	14709	14533

Gaussian mixture clustering metrics

Clustering Scores	Normalization by distance			Normalization by duration		
	PCA	SVD	None	PCA	SVD	None
CH ↑	**5602.8**	5596.9	5465.7	3585.2	3575.3	3517.7
DB ↓	**1.3**	**1.3**	**1.3**	1.6	1.7	1.7
S ↑	**0.38**	**0.38**	0.37	0.36	0.36	0.35
Instances cluster 0	5287	5301	5414	10594	10581	10419
Instances cluster 1	9715	9701	9588	4408	4421	4583

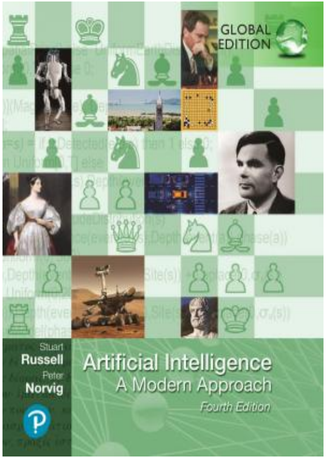
Identified driver styles

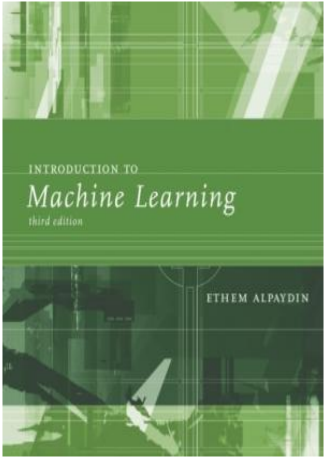
Label	Description	Number of instances
0	Non-Aggressive trips	11915
1	Aggressive trips	3034
2	Risky trips	53

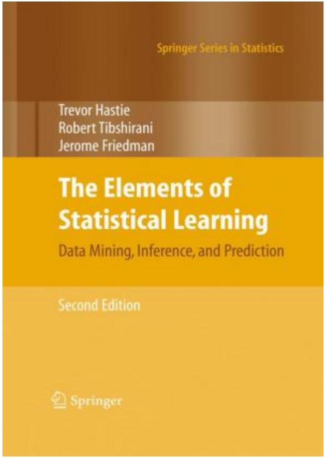
CONCLUSIONS

- Clustering algorithms find driver styles
- K-Means provides the best cluster separation
- Normalization of the trip data is necessary
  - by **distance** seems to be preferable than by **duration**

Bibliography - books about IA and ML







Bibliography - books about IA